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Retail Sales Data Acquisition, Preprocessing, and Exploratory Analysis

Introduction

The retail industry produces large volumes of transactional data every day. Each customer purchase, whether online or in-store, generates a digital record that contains valuable insights into consumer behavior, product performance, spending patterns, and operational trends. Analyzing this data effectively can provide retailers with a strong competitive advantage by supporting better decisions in inventory planning, pricing strategies, targeted marketing, and customer relationship management.

This report presents a comprehensive data acquisition and preprocessing analysis of a retail sales dataset. The dataset includes transaction-level details across various product categories, customer segments, payment methods, and geographic locations over a one-year period. The main objective of this assignment is to demonstrate a complete workflow for collecting, cleaning, transforming, and analyzing raw retail data, ultimately deriving meaningful and data-driven insights.

The analysis was carried out using Python and several key libraries, including Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, and Statsmodels. The notebook covers essential preprocessing techniques such as handling missing values, detecting and treating outliers, feature engineering, encoding categorical variables, scaling numerical features, applying PCA for dimensionality reduction, and performing exploratory data analysis through visualizations.

Problem Statement

Retail transaction data is often incomplete and noisy, with missing values and outliers that can reduce the reliability of the analysis. This study focuses on systematically acquiring, cleaning, and preprocessing retail sales data to enable accurate and meaningful insights. It examines key questions related to revenue-driving product categories and locations, spending behavior across time periods, payment methods, and the relationships between price, quantity, and total expenditure. Additionally, it explores the use of PCA (Principal Component Analysis) for dimensionality reduction while preserving variance. Overall, the work highlights the importance of robust data preprocessing in supporting reliable retail decision-making.

Methodology

The study follows a structured data analytics pipeline consisting of six key phases:

1. Data Acquisition

The retail sales dataset was loaded into a Pandas DataFrame using `read_csv()`. Initial inspection was performed using `data.info()` to understand data types and structure, and `data.isnull().sum()` to identify missing values across columns.

2. Data Cleaning

Missing values in numerical columns (Price Per Unit & Quantity) were imputed using column means, which is appropriate when skewness is approximately normal. The Total Spent column was recalculated as the product of Price Per Unit and Quantity to ensure consistency. The categorical Item column was filled with the mode (most frequent value). The Discount Applied column was cast to Boolean type and the missing entries were filled with False stating that they are treated as transactions with no discount.

3. Outlier Detection and Treatment

The Interquartile Range (IQR) method was employed to detect outliers in the numerical variables. The IQR is defined as $Q3 - Q1$, and the acceptable range is set to $[Q1 - 1.5 \times IQR \text{ for lower}, Q3 + 1.5 \times IQR \text{ for higher}]$. Values beyond these bounds were treated using the `clip()` function, which caps extreme values at the boundary thresholds without removing records, thereby preserving dataset integrity.

4. Exploratory Data Analysis (EDA)

EDA was conducted through groupby aggregations (monthly sales, category-wise sales, location-wise sales, top customers), a pivot table cross-tabulating category and payment method, correlation analysis, VIF computation for multicollinearity detection, and visualizations to uncover patterns in the data.

5. Feature Engineering

New features were created to enhance analysis, including Year, Month, Month Name, Day Name, and a Weekend indicator derived from transaction dates. A customer segmentation variable (Customer Segment) was created using spending bins with labels (Low, Medium, High, Premium). An Average Item Cost feature was computed as Total Spent divided by Quantity.

6. Dimensionality Reduction

StandardScaler was used to normalize the three numerical features before applying Principal Component Analysis (PCA) with two components. The explained variance ratio was used to evaluate how effectively the reduced components retained information.

Dataset Description

The dataset used in this study consists of 12,575 retail sales transactions with 11 attributes, where each row represents a unique purchase event. It contains both numerical and categorical variables, making it suitable for comprehensive exploratory and statistical analysis.

➤ **Transaction_ID:**

A unique identifier assigned to each transaction (e.g., TXN10000). It acts as a primary key and contains no missing values.

➤ **Customer_ID:**

An anonymized identifier for customers (e.g., CUST1346). Multiple transactions may belong to the same customer, enabling customer-level analysis and repeat purchase tracking.

➤ **Category:**

Represents the broad product category such as Furniture, Electric household essentials, Food, Milk Products, Butchers, Beverages, Computers and electric accessories and Patisserie etc. This is a nominal categorical variable.

➤ **Item:**

A coded representation of the specific product purchased in each transaction (e.g., Item_10_PAT, Item_17_MILK). The prefix indicates a unique item ID, while the suffix reflects the product category. This variable is categorical in nature and represents individual product-level details within broader categories.

➤ **Price_Per_Unit:**

The unit price of the product, recorded as a continuous numerical variable. It exhibited a right-skewed distribution and missing values were handled using mean imputation.

➤ **Quantity:**

The number of units purchased in the transaction, ranging from 1 to 10. This is a discrete numerical variable with an approximately normal distribution (skew = -0.063). Missing values were filled with the mean.

➤ **Total_Spent:**

Calculated as $\text{Price Per Unit} \times \text{Quantity}$ and represents the total transaction value. It serves as a key analytical variable and showed high right skewness due to extreme values, which were treated using IQR-based clipping.

➤ **Payment_Method:**

Indicates the mode of payment used, including Cash, Digital Wallet and Credit Card. This is a nominal categorical variable with no missing values.

➤ **Location:**

Represents the sales channel through which the transaction occurred, either Online or In-store. This categorical variable helps distinguish between digital and physical purchasing behavior.

➤ **Transaction_Date:**

The date of purchase covering a full year of transactions. It was converted to datetime format and used to derive Year, Month, Day Name, and Weekend features for temporal analysis.

➤ **Discount_Applied:**

A Boolean variable indicating whether a discount was applied (True/False). Missing values were treated as False, assuming no discount was given.

Data Acquisition Methods:

Data acquisition is the foundational step in any data analytics pipeline. It refers to the process of collecting, loading, and initially inspecting raw data before any transformation is applied. In this study, the following acquisition steps were followed:

- **Loading the Dataset**

The dataset was loaded from a CSV (Comma-Separated Values) file using the Pandas library function `pd.read_csv()`. This function reads the file into a DataFrame a two-dimensional tabular data structure that supports efficient manipulation, filtering, and aggregation. The initial load revealed a dataset with 12,575 records and 11 attributes.

- **Dataset Inspection**

The `data.info()` function was used to inspect the dataset, including column names, the count of non-null values per column, and the data type of each field. This step is critical because it immediately reveals which columns contain missing values and whether any column has been assigned an incorrect data type.

The inspection confirmed that Transaction ID, Customer ID, Category, Payment Method, and Location were complete had no missing records, while Item, Price Per Unit, Quantity, Total Spent, and Discount Applied contained missing entries.

- **Missing Value Quantification**

The `data.isnull().sum()` function was applied to quantify missing values across all columns. This step generates a column-wise count of null entries. Understanding the volume and location of missing data is essential for selecting appropriate imputation strategies in the data cleaning phase.

- **Duplicate Detection**

The `data.duplicated().sum()` function was applied to detect fully duplicated records where every value is identical to another row. Duplicate records can lead to inflated transaction counts, skewed revenue totals, and biased model training. The check confirmed that the dataset contained zero duplicate rows, verifying the uniqueness of every transaction record.

Analysis:

Refer Jupyter Notebook for complete analysis.

Conclusion:

This study successfully carried out a complete data acquisition and preprocessing workflow on the retail sales transaction dataset containing 12,575 records and 11 attributes. The process included dataset loading, structural inspection, missing value handling, skewness analysis, outlier detection and treatment, feature engineering, exploratory analysis, and dimensionality reduction. These steps transformed the raw dataset into a clean, reliable, and analysis-ready form suitable for further business intelligence and machine learning applications.

Missing values were handled using appropriate statistical techniques, while duplicate checks confirmed the uniqueness of all transaction records, ensuring high data integrity throughout the analysis. Outlier analysis using the IQR method revealed a small number of extreme values in the Total Spent column. Instead of removing records, clipping was applied to preserve transactional information while reducing skewness. This improved the overall data quality without affecting the dataset's central tendency.

Exploratory analysis provided valuable business insights. Butchers and Electricals emerged as the highest revenue generating category, while online store recorded strong sales performance in the location column. Monthly sales trends highlighted seasonal fluctuations, with January, July and December showing peak performance. Correlation analysis further showed that Total Spent is primarily influenced by Price Per Unit rather than Quantity, indicating that product pricing contributes more significantly to revenue generation.

Finally, Principal Component Analysis (PCA) demonstrated that most of the dataset's variability could be retained using few dimensions, confirming the efficiency of preprocessing. Overall, the study established a strong foundation for future predictive analytics tasks such as sales forecasting, customer segmentation, recommendation systems, and demand prediction, while also providing actionable insights for retail decision making and strategy development.

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